Artificial Intelligence: Search, Reasoning & Learning

01 / Search and games → 02 / Learning from data → 03 / Neural models → 04 / Probability and clusters

01 / Match the frontier to the objective

BFS minimizes depth; UCS minimizes accumulated nonnegative path cost. Equal positive step costs align them. Test a UCS goal when removed from the priority queue, and account for duplicate states.

02 / A* needs a justified heuristic

Use f=g+h. Admissibility is a lower bound; consistency also constrains every edge. Graph search with inconsistent estimates may need reopening. Beam truncation sacrifices ordinary A* guarantees.

03 / Optimization is budgeted exploration

Hill climbing can stop locally. Annealing accepts a worsening cost Δ with exp(-Δ/T). Genetic search needs valid encodings and diversity. A finite stochastic run is evidence of a found solution, not global optimality.

04 / Game backups

MAX takes maxima, MIN minima, chance nodes probability-weighted means. Alpha-beta prunes when alpha reaches beta for the same minimax tree. Heuristic cutoffs replace exact terminal guarantees with approximations.

05 / Prevent leakage

Fit preprocessing on training data only (refit per fold); make model choices on validation data. Keep the final test untouched. Use groups or time order when independent random splitting would misrepresent deployment.

06 / Evaluate the relevant errors

Precision=TP/(TP+FP), recall=TP/(TP+FN), F1=2TP/(2TP+FP+FN). Name the positive class, handle empty denominators and weight errors when their consequences differ.

07 / Tree splits and pruning

Information gain is parent entropy minus size-weighted child entropy. Unique identifiers can look excellent in training. Validate numeric thresholds, depth and pruning; gain ratio is a mitigation, not a guarantee.

08 / Backpropagation bookkeeping

Save a forward pass, compute all gradients with those parameter values, then update. For $L=\frac{1}{2}\Sigma(a-t)^2$ and sigmoid units, output $\delta=(a-t)a(1-a)$; hidden $\delta_j=a_j(1-a_j)\Sigma_k w_{kj}\delta_k$.

09 / Regularize deliberately

Learning rate affects stability; zero gradient need not indicate a minimum. Choose early stopping on validation. Inverted dropout scales retained training activations by 1/q (q = keep probability) and uses no dropout at inference.

10 / Margins and committees

Soft SVM balances norm against C-weighted slack; kernels must be valid inner products. Bagging reduces variance through resampling; boosting adapts emphasis; correlated errors limit the benefit of voting.

11 / Probability bookkeeping

Joint probabilities multiply local Bayesian-network factors. Sum hidden variables before normalizing. Conditioning differs from causation, and absence of a direct edge does not imply unconditional independence.

12 / Clustering choices change the answer

K-means pairs arithmetic means with squared Euclidean error. Single linkage can chain; complete linkage tests the farthest cross pair. Internal quality scores and cluster counts need interpretation and stability checks.